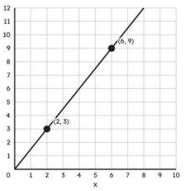
C-AR.4.1 · Weekly check · 5 questions

MA.7.AR.4.1 · Is it proportional? (table, graph, description) · Unit 11 Representing Proportional Relationships · Given Thursday, on the week's benchmark

Form title C-AR.4.1 Check · Is it proportional? (table, graph, description)
Description Learning goal: I can decide whether a relationship is proportional. 5 questions, 1 point each. Work each one out on paper, then choose or type your answer. Calculator allowed.
Locked mode OFF (calculator allowed; Locked mode blocks the Chromebook calculator)

#	Question image	File · type	Options to add	Answer key	Pts
1	Select **all** the situations that describe a proportional relationship. *Select **all** the correct answers.* A. A printer prints 30 pages every minute. B. A phone plan costs $36 plus $0.10 for each text message. C. A candle is 12 inches tall and burns down 1 inch each hour (its height compared with the hours burned). D. A taxi charges $3.00 plus $2.00 per mile. E. A store sells oranges at $2.40 per pound.	**Q1.png** Checkboxes	A B C D E	Check: **A, E** One point only if exactly these are checked.	1
2	Does the table show a proportional relationship between x and y? x | y 1 | 5 2 | 10 4 | 23 5 | 25 A. Yes, because $y \div x$ is the same for every pair of values. B. No, because the x-values do not increase by the same amount. C. No, because $y \div x$ is not the same for every pair of values. D. Yes, because the first two pairs of values have the same ratio.	**Q2.png** Multiple choice	A B C D	**C**	1
3	Does the graph show a proportional relationship between x and y?  A. Yes, because the line is straight and passes through the origin. B. Yes, because y increases whenever x increases. C. No, because the line does not reach the edge of the grid. D. No, because the line does not pass through the origin.	**Q3.png** Multiple choice	A B C D	**A**	1
4	A machine takes 15 minutes plus 10 minutes for each box packed. Is the relationship between the boxes packed and the total minutes proportional? A. Yes, because there is no setup time at the start, only 10 minutes each time. B. Yes, because the minutes added for each box packed never change. C. No, because there is a setup time of 15 minutes before the first box packed. D. No, because the minutes and the boxes packed are measured in different units.	**Q4.png** Multiple choice	A B C D	**C**	1
5	The table shows values of x and y. x | y 1.5 | 1.8 2 | 2.4 2.5 | 3 5 | 6 Noor says the relationship cannot be proportional because the table does not include $x = 0$. Is Noor's reason correct? Is the relationship proportional, and if it is, what is the constant of proportionality? A. Noor's reason is not correct; the relationship is proportional, with constant of proportionality 1.2. B. Noor's reason is correct; a proportional table must include $x = 0$, so the relationship is not proportional. C. Noor's reason is not correct, but the relationship is not proportional, because the x-values do not increase by the same amount. D. Noor's reason is not correct; the relationship is proportional, with constant of proportionality 0.6.	**Q5.png** Rigor Multiple choice Same problem, now multiple choice (paper: a written answer)	A B C D	**A**	1
6	LEARNING GOAL **I can decide whether a relationship is proportional.** 4 I can explain why a relationship is not proportional using more than one representation. 3 I can decide whether a table, graph, or written description shows a proportional relationship and say why. 2 I can check whether two ratios are equal. 1 I need help to start this.	**Level scale.png** Multiple choice (the template's last question)	Level 1 Level 2 Level 3 Level 4	no key	0

Answers with worked solutions are in the Answers folder next to the question images (same file names).